Ch-5

1) Demonstrate the various error detection techniques at data link layer with example.

Error detection techniques at the Data Link Layer are crucial for ensuring the integrity of data as it is transmitted over a network. These techniques help identify errors that may have occurred during transmission due to noise, interference, or other issues.

1. Parity Check

Description: Parity check is the simplest error detection method. It adds a parity bit to the data to make the number of 1s either even (even parity) or odd (odd parity).

Types:

- **Even Parity**: The number of 1s in the data plus the parity bit should be even.
- **Odd Parity**: The number of 1s in the data plus the parity bit should be odd.

Example:

- Data: `1011001` (contains 4 ones)
- Even Parity: Add a `0` → `10110010`
- Odd Parity: Add a `1` → `10110011`

Error Detection:

- If the received data `10110010` is checked and has an even number of 1s, it's accepted. If it has an odd number, an error is detected.

2. Checksum

Description: A checksum is a value calculated from a data set, which is sent along with the data. The receiver calculates the checksum again and compares it with the received checksum.

Example:

- Suppose we have the data segments: `10011010`, `10101100`, `11010101`.
- **Calculate the Checksum**:
  - First, convert the binary numbers to decimal and sum them:
    - `10011010` → 154
    - `10101100` → 172
    - `11010101` → 213
  - Total Sum: 154 + 172 + 213 = 539.
- **Convert to Binary**:
  - 539 in binary is `1000010111`.
  - Keep the last 8 bits for the checksum: `00001011`.

Sending:

- Data: `10011010`, `10101100`, `11010101` with Checksum: `00001011`.

Error Detection:

- The receiver calculates the checksum of the received data and checks against the received checksum. If they don't match, an error is detected.

3. **Cyclic Redundancy Check (CRC)**

Description: CRC is a more complex error detection method that uses polynomial division to detect changes to raw data. A fixed-length sequence (CRC code) is appended to the data.

Example:

- Let's say we want to send the binary data `1101011011` and use a polynomial `1101` (which represents `x^3 + x^2 + 1`).

Steps:

1. **Append Zeros**: Append three zeros to the data (`1101011011000`).
2. **Perform Polynomial Division**:
   - Divide `1101011011000` by `1101`. The remainder after the division is the CRC code.
3. **Send Data**:
   - The transmitted data becomes `1101011011` + CRC remainder.

Error Detection:

- The receiver performs the same division. If the remainder is zero, there is no error; if not, an error is detected.

Hamming Code

Description: Hamming code not only detects errors but can also correct single-bit errors. It adds redundant bits to the data at specific positions.

Example:

- For a 4-bit data `1011`, we want to send it using Hamming (7,4) code.
- Positions of parity bits: 1, 2, and 4 (P1, P2, P4), data bits at positions 3, 5, 6, 7.

Data with Parity Bits:

- Initial positions: `P1 P2 D1 P4 D2 D3 D4`
- Fill in the data: `P1 P2 1 P4 0 1 1`

Calculate Parity Bits:

- P1 checks bits: 1, 3, 5, 7
- P2 checks bits: 2, 3, 6, 7
- P4 checks bits: 4, 5, 6, 7

Sending:

- After calculating the parity bits, the complete code word is sent.
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Error Detection and Correction:

- The receiver checks the parity bits and can identify which bit (if any) is incorrect and correct it

2) Explain the following static channel allocations mechanisms: i) TDMA ii) FDMA

Static channel allocation mechanisms are methods used in communication networks to allocate bandwidth among multiple users or devices. Two common static channel allocation techniques are **Time Division Multiple Access (TDMA)** and **Frequency Division Multiple Access (FDMA)**. Here's a detailed explanation of each:

1. Time Division Multiple Access (TDMA)

Description: TDMA is a channel access method that divides the available bandwidth into time slots. Each user or device is assigned a specific time slot during which they can transmit their data. This method allows multiple users to share the same frequency channel without interfering with each other, as they transmit in different time intervals.

How TDMA Works:

- **Time Slots**: The total available time is divided into fixed time slots. Each user is allocated a specific time slot in a round-robin fashion.
- **Synchronization**: All users must synchronize their clocks to ensure they transmit in their assigned slots.
- **Static Allocation**: Time slots are typically allocated statically, meaning each user has a fixed time slot for transmission.

Example:

Imagine a scenario where three users (A, B, and C) share a communication channel using TDMA, with time slots of 1 ms each:

- **Time Slot Allocation**:
  - **A**: 0-1 ms
  - **B**: 1-2 ms
  - **C**: 2-3 ms
  - **A**: 3-4 ms
  - And so on...

Advantages of TDMA:

- Efficient use of bandwidth as it avoids collisions.
- Predictable performance, which is beneficial for real-time applications.

Disadvantages of TDMA:

- Requires precise synchronization among users.
- Inefficient in scenarios where users have variable data transmission requirements, as idle slots can occur.

2. Frequency Division Multiple Access (FDMA)

Description: FDMA is another channel access method that divides the available bandwidth into distinct frequency bands. Each user is assigned a specific frequency band for transmission, allowing multiple users to transmit simultaneously without interference.

How FDMA Works:

- **Frequency Bands**: The entire frequency spectrum is divided into smaller frequency bands. Each user is allocated a unique frequency range.
- **Static Allocation**: Frequency bands are typically allocated statically, meaning each user has a fixed frequency band for transmission.

Example:

Consider a scenario where three users (A, B, and C) share a communication channel using FDMA, with the frequency range divided as follows:

- **Frequency Band Allocation**:
  - **A**: 2.4 GHz to 2.5 GHz
  - **B**: 2.5 GHz to 2.6 GHz
  - **C**: 2.6 GHz to 2.7 GHz

In this example, each user can transmit their data simultaneously, as they are using different frequency bands.

Advantages of FDMA:

- Allows simultaneous transmission by multiple users, improving overall throughput.
- Simple to implement with minimal synchronization requirements.

Disadvantages of FDMA:

- Inefficient use of bandwidth, as users may not fully utilize their allocated frequency bands.
- Frequency bands must be carefully managed to avoid interference between users.

Comparison of TDMA and FDMA

Feature	TDMA	FDMA
Basic Concept	Time slots allocation	Frequency bands allocation
Transmission	Sequential (one user at a time)	Simultaneous (multiple users)
Synchronization	Required for time slots	Minimal required
Bandwidth Utilization	More efficient in constant data	Can be inefficient
Complexity	More complex due to synchronization	Simpler implementation
Use Cases	Suitable for real-time applications	Used in traditional telephony

3) Explain p-persistent CSMA protocol in detail.

p-Persistent CSMA Protocol

Description: p-Persistent Carrier Sense Multiple Access (CSMA) is a network protocol used in environments where multiple devices share a communication channel. This protocol is an improvement over traditional CSMA methods, where devices listen to the channel before transmitting data. In p-persistent CSMA, devices transmit with a certain probability, p, after detecting that the channel is idle.

How p-Persistent CSMA Works:

1. **Carrier Sensing**: Before attempting to send data, a device first listens to the channel to check if it is idle (i.e., no other devices are transmitting).
2. **Probability Decision**:
   - If the channel is idle, the device transmits with a probability p
   - If the device does not transmit (with a probability 1−p), it waits for the next time slot and listens again.
3. **Collision Handling**:
   - If two devices transmit simultaneously (which can happen if they both decide to transmit after sensing an idle channel), a collision occurs.
   - When a collision is detected (usually by a lack of acknowledgment), each device waits a random amount of time before attempting to transmit again.
4. **Exponential Backoff**: In many implementations, devices use an exponential backoff strategy after a collision to reduce the chances of repeated collisions.

Flow of Events in p-Persistent CSMA:

- **Step 1**: Device A wants to transmit data.
- **Step 2**: Device A senses the channel:
  - If the channel is idle, it generates a random number:
    - If this random number is less than or equal to p, it transmits.
    - If it's greater than p, it waits for the next time slot.
- **Step 3**: If Device A transmits, it monitors for collisions. If no collision occurs, the transmission is successful. If a collision is detected, it will wait for a random period and attempt to retransmit later.

Advantages of p-Persistent CSMA:

- Reduces the probability of collisions compared to 1-persistent CSMA (where devices always transmit when the channel is idle).
- Flexible parameter p
- p allows for tuning the protocol to suit the specific traffic conditions of the network.

Disadvantages of p-Persistent CSMA:

- The effectiveness of p-persistent CSMA decreases under high traffic conditions, as the probability of collisions increases.
- Requires careful tuning of p
- p to optimize performance for specific network environments.

Comparison with Other CSMA Variants

- **1-Persistent CSMA**: Transmits immediately when the channel is idle, leading to higher collision rates.
- **Non-persistent CSMA**: Waits until the channel is idle but may lead to under-utilization of the channel as devices do not sense the channel continuously.
- **p-Persistent CSMA**: Balances between these two extremes, allowing for a controlled approach to transmission based on a probability p

4) Data link protocols almost always put CRC in a trailer rather than in a header. Why?
5) State the difference between bit rate and baud rate.
6) Discuss the working of slotted aloha along with its efficiency in terms of channel utilization.

Slotted ALOHA is a protocol used for random access in communication networks, particularly in scenarios where multiple users need to share a communication channel. It is an improvement over the original ALOHA protocol, which allows for better efficiency in channel utilization.

Working of Slotted ALOHA

Basic Principles:

1. **Time Division**: In Slotted ALOHA, time is divided into discrete slots of equal length. Each slot corresponds to the time it takes to send one packet of data.

2. **Synchronization**: All users in the network must synchronize their clocks to ensure that they start transmitting at the beginning of a time slot. This synchronization is crucial as it prevents collisions that could occur if users transmit at arbitrary times.

3. **Transmission Process**:

    - When a user has data to send, they wait for the beginning of the next time slot.
    - The user then transmits their data at the start of the time slot.
    - If the packet is successfully transmitted without any collisions, the user receives an acknowledgment (ACK).
    - If a collision occurs (i.e., two or more users transmit during the same time slot), the data packets become garbled and are not successfully received. In this case, the users will wait for a random amount of time before attempting to retransmit, often using a random backoff algorithm.
4. **Collision Detection and Retransmission**: After a collision, users will not receive an acknowledgment and will assume that the transmission failed. They will then randomly choose a time slot in the future to retransmit their data.

Efficiency of Slotted ALOHA

The efficiency of a protocol can be measured in terms of channel utilization, which refers to the proportion of time that the channel is effectively used for transmitting data without collisions.

This means that the maximum channel utilization (or efficiency) of Slotted ALOHA is approximately **36.8%**.

Comparison with Pure ALOHA:

- **Efficiency**: Pure ALOHA has a maximum efficiency of about 18.4% (when G=0.5), which is half that of Slotted ALOHA.
- **Collision Time**: In Pure ALOHA, collisions can occur at any time, leading to a higher chance of overlapping transmissions. In Slotted ALOHA, collisions can only occur at the beginning of time slots, reducing the chance of collisions and thus increasing efficiency.

7) Explain different types of CSMA with suitable example.

Carrier Sense Multiple Access (CSMA) is a network protocol that manages how multiple devices access a shared communication medium. It uses a "listen before talk" approach, where devices first check the channel to see if it's free before transmitting data. There are several variations of CSMA, each with its own method for handling access and collisions.

1. 1-Persistent CSMA

Description: In 1-persistent CSMA, a device continuously senses the channel and transmits immediately when it detects that the channel is idle. If the channel is busy, the device keeps listening until it becomes free and then sends its data immediately.

Example:

- Imagine three devices (A, B, and C) that want to transmit data. If Device A senses that the channel is idle, it transmits its data. If Device B is also waiting to transmit and senses the channel is idle immediately after Device A starts sending, Device B will wait until Device A finishes. However, if both devices sense the channel at the same time and it is idle, they may collide when they both transmit simultaneously.

Advantages:

- Simple to implement.
- Low latency in the absence of collisions.

Disadvantages:

- High collision rate in a busy network, leading to reduced efficiency.

Non-Persistent CSMA

Description: In non-persistent CSMA, when a device finds the channel busy, it does not keep listening. Instead, it waits a random period before checking the channel again. This reduces the

chance of collisions, as devices do not immediately attempt to transmit when the channel becomes free.

Example:

- In the same scenario with Devices A, B, and C, if Device A senses the channel and finds it busy, it will wait a random time before checking again. If the channel is free, it will then transmit. This method reduces the likelihood that Devices B and C will collide when they try to transmit after A.

Advantages:

- Reduced collision rates compared to 1-persistent CSMA.

Disadvantages:

- Potentially longer delays for transmission due to random waiting.

p-Persistent CSMA

Description: p-persistent CSMA is a probabilistic approach where a device transmits with a probability p when it detects that the channel is idle. If it decides not to transmit, it waits until the next time slot to sense the channel again.

Example:

- If the channel is idle, Device A has a 70% chance (p=0.7) of transmitting its data. If it does not transmit, it waits for the next time slot. If Devices A and B are both waiting to transmit and find the channel idle, each has a chance to decide independently whether to send or wait.

Advantages:

- Balances efficiency and collision avoidance by reducing transmission frequency based on probability.

Disadvantages:

- Requires tuning the probability p for optimal performance.

CSMA/CD (Carrier Sense Multiple Access with Collision Detection)

Description: CSMA/CD is an extension of the basic CSMA protocol that incorporates collision detection. Devices continue to listen while transmitting. If a collision is detected (indicated by a sudden change in signal), the devices stop transmitting and initiate a backoff strategy before attempting to resend their data.

Example:

- Devices A, B, and C are all transmitting on the same channel. If A and B transmit at the same time, they both detect the collision and stop transmitting. They then wait a random time before trying to transmit again, which reduces the chances of repeated collisions.

Advantages:

- More efficient than pure CSMA due to reduced collisions and better handling of traffic.

Disadvantages:

- More complex than basic CSMA.

- Not effective in wireless environments where collision detection is more challenging.

CSMA/CA (Carrier Sense Multiple Access with Collision Avoidance)

Description: CSMA/CA is primarily used in wireless networks. It avoids collisions by using a mechanism to prevent them from happening in the first place. Devices send a request to send (RTS) message before transmitting data, and the recipient responds with a clear to send (CTS) message, allowing the sender to transmit.

Example:

- In a wireless network, Device A wants to send data to Device B. It first sends an RTS to Device B. If Device B is ready to receive, it replies with a CTS. Device A then transmits the data. Other devices that hear the RTS or CTS will defer their transmissions, reducing the likelihood of collision.

Advantages:

- Reduces collision probability in wireless networks.
- Allows better management of bandwidth in shared wireless environments.

Disadvantages:

- Higher overhead due to RTS/CTS exchanges.
- Increased latency due to waiting for acknowledgments.

CSMA Type	Mechanism	Advantages	Disadvantages
1-Persistent	Transmits immediately when idle	Low latency, simple	High collision rates in busy networks
Non-Persistent	Waits a random time if busy	Reduces collisions	Longer delays due to random waiting
p-Persistent	Transmits with a probability p	Balances efficiency and collision avoidance	Requires tuning of p
CSMA/CD	Detects collisions during transmission	More efficient with lower collisions	More complex; not suitable for wireless
CSMA/CA	Avoids collisions using RTS/CTS	Reduces collision in wireless networks	Higher overhead and increased latency

8) Generate CRC code for message 1101010101. Given generator polynomial is x4+x2+1.